Technical University of Munich
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Operations Research

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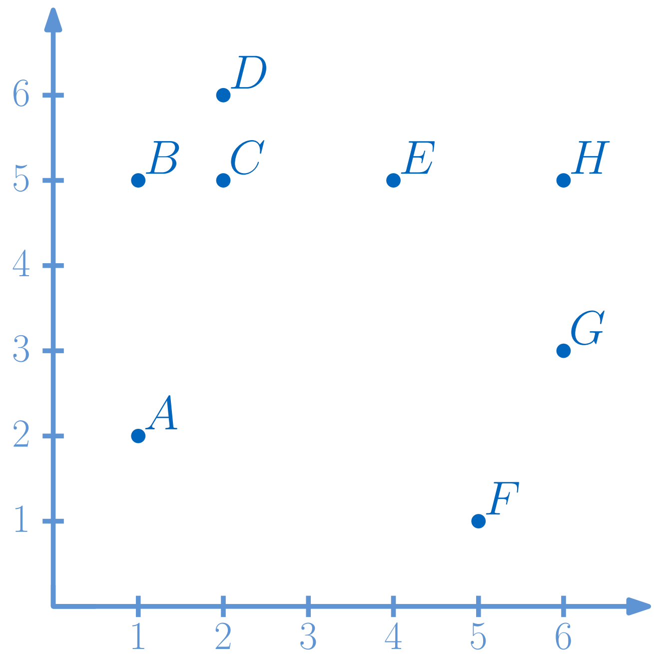
Sample solution to the demo tasks 9

Demo Exercises

Exercise D9. 1. *TSP-Approximation*

Problem

Consider the points A to H in the plane:



The distance between $p_1 = (x_1, y_1) \in \mathbb{R}^2$ and $p_2 = (x_2, y_2) \in \mathbb{R}^2$ is given by the euclidean norm: $d(p_1, p_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

a) Determine a minimum spanning tree using Kruskal's algorithm. List the edges in the order they are added by the algorithm.

b) Provide the tour you obtain when you execute the 2-approximation algorithm of Christofides using your spanning tree from part (a). Start your tour at node A.

c) What is the length of the tour you obtained and which lower bound can you give for the optimal tour? Where does the approximation factor of 2 come from?

Solution

a) The order of the nodes in specifying an edge is not important in this task, i.e., $AB = BA$.

Distance	Pairs
1.00	BC, CD
1.41	BD
2.00	CE, EH, GH
2.24	DE, FG
2.83	EG
3.00	AB, BE
...	...

The algorithm processes the edges in ascending order of their distances. It begins with BC and CD. Adding these to the forest I does not violate the cycle-freeness condition, so both are included. However, BD is excluded because it would create a cycle with BC and CD. Next, the edges CE, EH, and GH are considered. Since they also do not form cycles, they are added in any order. Although DE follows, it is skipped due to cycle formation. The edge FG is added without issue. Finally, AB is included, bringing the total to seven edges — the maximum possible in an MST for eight vertices. One valid edge inclusion order is BC, CD, EH, GH, CE, FG, AB, but this is not unique. Several permutations lead to the same (in this case unique) MST.

b) Depending on the direction in which the MST is traversed, different solutions are obtained. A distinction is made between the "inside" and "outside" traversal, resulting in the following tours:

- Inside: $A \to B \to C \to E \to H \to G \to F \to D \to A$
- Outside: $A \to B \to C \to D \to E \to H \to G \to F \to A$

c) The length of the Minimum Spanning Tree is 13.24. The corresponding tour lengths are 17.6 (outside) and 22.19 (inside). Using Christofides' algorithm, which provides a 2-approximation for the Travelling Salesman Problem (TSP), we obtain three lower bounds for the optimal tour:

- $22.19/2 = 11.09$ from the inside path
- $17.6/2 = 8.8$ from the outside path
- 13.24, the actual MST length, which is itself a lower bound for the TSP tour

Thus, the optimum lies between 13.24 and 17.6 (with 17.6 being the actual optimum in this case).

The approximation factor of 2 stems from the fact that Christofides' algorithm doubles each edge of the MST to create an Eulerian graph before shortcutting. Therefore, the Christofides tour has a length of at most twice the optimum.

Exercise D9. 2. *Collect electric scooters*

Problem

Fabienne and Mattea work for the electric scooter operator *boi* and are tasked with collecting empty scooters and taking them to a depot. Fabienne is stationed in depot 1 (D_1), while

Mattea is in depot 2 (D_2). The two are given a list of locations $S_1, \ldots, S_n$ with empty scooters that they should collect ($n \geq 4$). The cost of a trip from node i to node j for $i, j \in V$ is denoted by c_{ij}. Let $V = \{S_1, \ldots, S_n, D_1, D_2\}$ be the set of all nodes and let the variables x_{ij}^F or x_{ij}^M be whether Fabienne or Mattea uses the line segment between $i, j \in V$ for $i \neq j$. Fabienne and Mattea want to model the problem at hand as an integer linear program and determine the minimum effort for both.

Formulate the following constraints. If you introduce new variables, also provide a brief description.

a) Each scooter location must be visited exactly once.

b) Fabienne and Mattea leave their respective depots exactly once and also return exactly once.

c) Mattea and Fabienne each do a round trip.

d) Finally, formulate an appropriate objective function.

Solution

a)

$$\sum_{i \in V, i \neq j} x_{ij}^F + x_{ij}^M = 1 \, \forall j \in V \setminus \{D_1, D_2\}$$

b) Fabienne and Mattea leave their depot exactly once.

$$\sum_{i \in V,\, i \neq D_1} x_{D_1 i}^F = 1 = \sum_{i \in V,\, i \neq D_2} x_{D_2 i}^M$$

Fabienne and Mattea return to their depot exactly once.

$$\sum_{i \in V,\, i \neq D_1} x_{i D_1}^F = 1 = \sum_{i \in V,\, i \neq D_2} x_{i D_2}^M$$

c) For a round trip, each node (including D_1 and D_2) must be visited as often as it is left:

$$\sum_{i \neq j} x_{ij}^k = \sum_{i \neq j} x_{ji}^k \, \forall j \in V \, \forall k \in \{F, M\}$$

We must now - similar to the TSP - prevent sub-tours from being created that do not contain D_1 or D_2. This can be done with the SEC constraints, for example:

$$\sum_{i,j \in U_F,\, i \neq j} x_{ij}^F \leq |U_F| - 1 \, \forall U_F \subseteq V \setminus \{D_1\},$$

$$\sum_{i,j \in U_M,\, i \neq j} x_{ij}^M \leq |U_M| - 1 \, \forall U_M \subseteq V \setminus \{D_2\}.$$

Alternatively, the MTZ constraints can also be used:

$$u_{D_1}^F = 1 \tag{1}$$
$$u_i^F \geq 2, \text{ für } i \in V \setminus D_1 \tag{2}$$
$$u_i^F \leq |V|, \text{ für } i \in V \tag{3}$$
$$u_i^F - u_j^F + 1 \leq (|V| - 1)(1 - x_{ij}^M), \text{ für } i, j \in V, j \neq D_1. \tag{4}$$

The same applies to Mattea:

$$u^M_{D_2} = 1 \tag{5}$$
$$u^M_i \geq 2, \text{ für } i \in V \setminus D_2 \tag{6}$$
$$u^M_i \leq |V|, \text{ für } i \in V \tag{7}$$
$$u^M_i - u^M_j + 1 \leq (|V| - 1)(1 - x^M_{ij}), \text{ für } i, j \in V, j \neq D_2. \tag{8}$$

d) They want to minimize their effort, so the objective function is as follows:

$$\min \sum_{i,j \in V, i \neq j} c_{ij}(x^F_{ij} + x^M_{ij}).$$

Exercise D9. 3. *NP-hardness of TSP*

Problem

The decision problem of whether a Hamiltonian cycle (HC) exists in an undirected and unweighted graph $G = (V, E)$ is generally NP-complete. Show that the *metric Traveling Salesperson Problem* (TSP) is at least as hard as the HC problem.

To do this, show that you can represent any HC problem as a solution to a specific TSP. In doing so, you will have demonstrated a *reduction* from the HC problem to the TSP problem.

Solution

Given an undirected graph $G = (V, E)$, we first extend the graph with weights and additional edges. Denote by $E_{\text{all}} := \{(i, j) | i, j \in V \text{ for } i \neq j\}$ the set of all possible edges between all nodes in V. Let $E' = E_{\text{all}} \setminus E$ be the set of all edges that are not yet included in the graph G. Note that $E' \cup E = E_{\text{all}}$.

We set the edge weights as follows:

$$u(i, j) = \begin{cases} 1, \text{ for } (i, j) \in E \\ 2, \text{ for } (i, j) \in E' \end{cases} . \tag{9}$$

See the schematic representation of the transformation in the given figure.

The solution to the TSP decision problem for the weighted graph $G' = (V, E_{\text{all}}, u)$ - "Does a path of length $|V|$ exist?" - precisely solves the HC problem.

If the answer is yes, we must have found a path of TSP that visits each node exactly once, ends at the starting node, and only uses edges of weight 1. If it used any edge of weight 2, the length would have been $> |V|$. So, a solution of length $|V|$ to the TSP problem is a solution to the original HC problem. On the other hand, if the answer is no, then any TSP solution must use an edge of length 2, and such an edge is not present in the original HC problem, so no HC solution exists. Thus, we have shown that every HC problem can be solved by our reformulated TSP. Consequently, the TSP is at least as hard as the HC problem. It follows that the TSP is NP-hard.

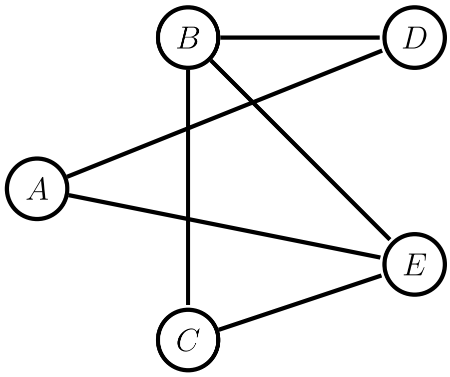


Figure 1: Graph G - Does there exist a hamiltonian cycle?

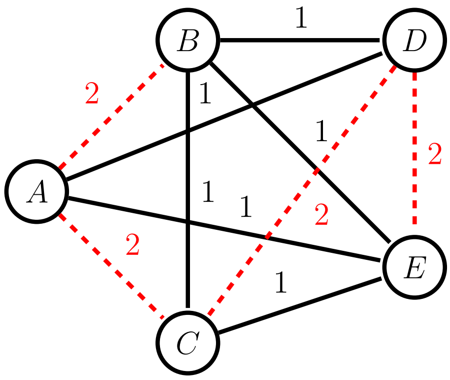


Figure 2: TSP formulation - Does there exist a path with length n?